

Survey Research and Design in Psychology

Tutorial 2 - Correlational analysis

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- ▶ Talk about the general steps to take when running a correlation
- ▶ Go over the types of correlation and level of measurement
- ▶ Practice running correlations, scatterplots
- ▶ Look at the effect of outliers, restricted ranges
- ▶ Correlations and non-linear relations

What is a correlation?

- ▶ A standardised measure of how related two variables are
- ▶ It ranges from -1 to $+1$, with more extreme values indicating a stronger relationship
- ▶ We are asking: what is the relationship/association/shared variance between two variables

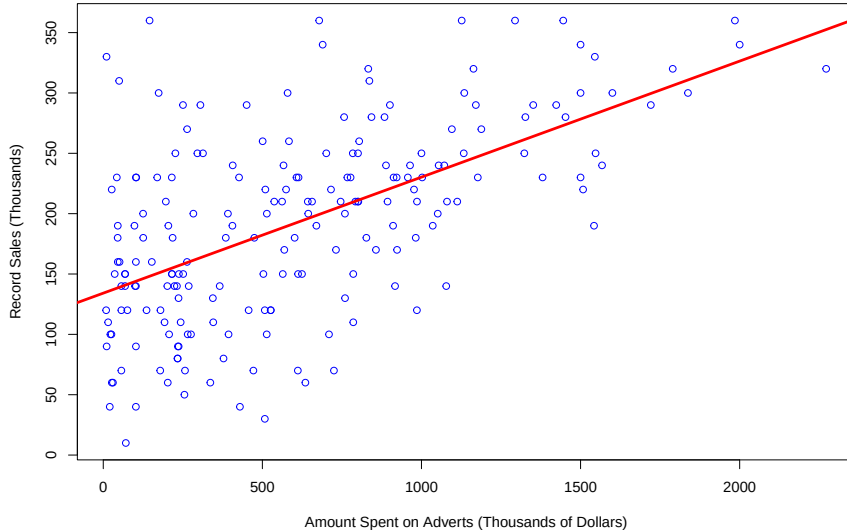
- ▶ To interpret a linear correlation, look at the:
 - ▶ Direction - is the relationship positive or negative? Check the (+/-)
 - ▶ Strength - how strong is the relationship? Check the size (-1 to + 1)
 - ▶ Statistical significance - is there a significant relationship? Check the p-value (<.05 indicates a significant relationship)

Linear correlation

```
##
## Call:
## lm(formula = sales ~ 1 + adverts, data = album_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -152.949  -43.796   -0.393   37.040  211.866
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.341e+02  7.537e+00  17.799  <2e-16 ***
## adverts      9.612e-02  9.632e-03   9.979  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 65.99 on 198 degrees of freedom
## Multiple R-squared:  0.3346, Adjusted R-squared:  0.3313
## F-statistic: 99.59 on 1 and 198 DF,  p-value: < 2.2e-16
```

Linear correlation

Advertisement Investment and Number of Records Sold



General steps in a correlation analysis

- ▶ Determine the level of measurement for each variable
- ▶ Examine univariate descriptive statistics and graphical displays for each variable
- ▶ Recode if necessary
- ▶ Examine a bivariate graph
- ▶ Examine correlational statistics
- ▶ Interpret and make conclusions

- ▶ Nominal - placing cases into named categories
- ▶ Ordinal - placing cases in order (ranking them)
- ▶ Interval/Ratio - placing cases on a scale with meaningful distances between points (ratio has a 0 point)

Types of correlation

- The type of correlation you should use depends on the level of measurement of your variables

	Nominal	Ordinal	Interval/Ratio
Nominal	Chi-squared (X^2), Phi or Cramer's V, Clustered bar chart	← Treat as for	Point bi-serial correlation (r_{pb}), Scatterplot, bar chart, or error-bar chart
Ordinal		Spearman's Rho (r_s) or Kendall's Tau, Clustered bar chart OR scatterplot	Treat as for ↑ and ←
Interval/ Ratio			Pearson's Product-moment correlation (r), Scatterplot

1. Frequency tables and crosstabs

- ▶ Is there enough data in each category?
- ▶ If not, it might be necessary to recode the data into fewer categories

2. Clustered bar chart

- ▶ Frequencies or percentages on the y-axis?

3. Inferential test

- ▶ Pearson chi-square test

4. Effect size

- ▶ Phi (ϕ) for 2x2, 2x3, or 3x2 tables
- ▶ Cramer's V for tables 3x3 and larger
- ▶ Non-parametric, but check for a minimum expected frequency of 5 per cell
- ▶ Note - be careful interpreting the sign

Download Tutorial 2 data

Tutorial 2 data

Familiarise yourself with the data

- ▶ How many participants?
- ▶ What is the level of measurement of the smoking variable?
- ▶ How many levels does the belief in God variable have?
- ▶ How is gender coded? Gender role?

- ▶ What is the relationship between gender and belief in god?
- ▶ To answer this question, we need to follow the following steps:
 - ▶ Univariate frequencies and crosstabs
 - ▶ Clustered bar chart
 - ▶ Pearson's chi-square test
 - ▶ Effect size

Univariate frequencies and bar charts

- ▶ Which statistics do we request?
 - ▶ The default includes mean and median. Is this sensible?
 - ▶ Select Frequency tables
- ▶ What kind of plot is most appropriate?
 - ▶ Select Bar plot
 - ▶ Notice anything strange in the God variable? Inspect this further.

- ▶ Recode God variable to remove error
 - ▶ Select the God variable > click transform > create new transform
 - ▶ Create a name, description and suffix (e.g. cln) for the transformation, which will recode the 3's to empty values
 - ▶ Select Add recode condition
 - ▶ if \$source <= 2 use \$source else use
 - ▶ Examine barchart for recoded God variable

Crosstabs, chi-square test, effect size

- ▶ Analyses > frequencies > independent samples contingency table
- ▶ Move transformed God variable into rows
- ▶ Move Gender into columns
- ▶ Statistics
 - ▶ Tests - Chi-Square
 - ▶ Nominal - Phi and Cramer's V
- ▶ Cells
 - ▶ Counts
 - ▶ Observed counts
 - ▶ Expected counts
 - ▶ Percentages
 - ▶ Column

- ▶ A tip for interpreting the output
 - ▶ The % within the column is the percentage of people in each gender category
 - ▶ 0 = Yes, 1 = No, 2 = Sort of

Crosstabs, chi-square test, effect size

- ▶ Is there a relationship between gender and belief in God?

Clustered bar graph

- ▶ Analyses > Exploration > Descriptives
- ▶ Move God - cln into the variables box
- ▶ Split by Gender
- ▶ Select bar plot

We conducted a chi-square analysis to examine whether there was a relationship between gender (male vs female) and belief in God (yes, no, sort of). This analysis showed no significant difference in belief in god across genders ($\chi^2(2, 184) = 0.11$, $p = .949$), with a negligible effect size as indicated by Cramer's V (.02), suggesting that men and women are just as likely to believe - and disbelieve - in God.

- ▶ What is the relationship between smoking and snoring?
 - ▶ Univariate frequencies and crosstabs
 - ▶ Clustered bar chart
 - ▶ Pearson's chi-square test
 - ▶ Effect size

- ▶ Analyses > Exploration > Descriptives
- ▶ Move Smoking and Snoring into the variables box and request frequency tables and histograms
- ▶ This will give you the frequencies for snoring and the histogram for smoking. Why?
 - ▶ The type of variable will influence what is returned
 - ▶ To graph snoring appropriately, request a bar plot. Histogram will only return a meaningful graph for smoking.

Univariate frequencies and bar charts

- ▶ Let's recode the smoking variable to a nominal variable (i.e. smoking - never/ever)
- ▶ To do this, we will need to create another transform function

Recode smoking into two categories

- ▶ Select the smoking variable > click transform > create new transform
- ▶ Enter the details of the transform (name i.e. "transform_to_categorical", description i.e. "Recode into 0 (never) to 1 (ever)", suffix i.e. "bin") > Add recode condition > if \$source == 0 use 0 else use 1
- ▶ Check your work - request a bar plot of Smoking-bin
 - ▶ Notice, our new smoking-bin variable is automatically given a continuous measure type
 - ▶ Go back into this transformation to change this to nominal (i.e. Measure type = Nominal)

What is the relationship between smoking and snoring

- ▶ Let's now eyeball this relationship by requesting the clustered bar chart
 - ▶ Analyses > Exploration > Descriptives > Smoking - bin in variables and snoring in split by > bar plot

What is the relationship between smoking and snoring

- ▶ Frequencies > Independent samples Chi-square test > Smoking - bin in Rows and Snoring in Columns
- ▶ Statistics
- ▶ Tests
 - ▶ χ^2
- ▶ Comparative measures (2x2 only)
 - ▶ Confidence intervals - Interval 95%
- ▶ Nominal
 - ▶ Phi and Cramer's V
- ▶ Cells
 - ▶ Counts
 - ▶ Observed counts
 - ▶ Expected counts
 - ▶ Percentages
 - ▶ Row

What is the relationship between smoking and snoring

- ▶ What pattern do you see in the contingency table?
- ▶ Is the chi-square test indicating significance?
- ▶ How much of an effect is there?

We were interested in whether there was a relationship between smoking and snoring. To test this, we used a chi-square analysis, which revealed that smokers are more likely to snore than nonsmokers ($\chi^2(1, 188) = 8.07, p = .004, \phi = .21$), with the Phi coefficient indicating a small effect size.

Types of correlation

► Nominal x interval/ratio

	Nominal	Ordinal	Interval/Ratio
Nominal	Chi-squared (χ^2), Phi or Cramer's V, Clustered bar chart	← Treat as for	Point bi-serial correlation (r_{pb}), Scatterplot, bar chart, or error-bar chart
Ordinal		Spearman's Rho (r_s) or Kendall's Tau, Clustered bar chart OR scatterplot	Treat as for ↑ and ←
Interval/ Ratio			Pearson's Product-moment correlation (r), Scatterplot

Dichotomous by interval/ratio

- ▶ Use the point biserial correlation (r_{pb})
- ▶ Graph using a scatterplot, bar chart, or error-bar chart
- ▶ Compute product-moment correlation (r)
- ▶ Interpret the r (hint: think about how your dichotomous variable is coded!)

- ▶ What is the relationship between Gender and Australianness?
 - ▶ Univariate frequencies and bar graphs
 - ▶ Generate a scatterplot
 - ▶ Run a correlation
 - ▶ Effect size

Univariate frequencies and bar charts

- ▶ Analyses > Exploration > Descriptives
- ▶ Move Gender and Australianness into the variables box.
 - ▶ For these variables, let's just take a quick look at them graphically
 - ▶ Note: Check their measurement type and then choose the appropriate plots (i.e. histogram - Australianness and bar plot - Gender)

Generate scatterplots

- ▶ Jamovi does not have an option to run scatterplots in the base version
- ▶ We need to install a module from the jamovi library called scatter
- ▶ Select Modules > jamovi library > scatr (I'm using version 1.2.0)

Generate scatterplots

- ▶ We can now graph the correlation in scatr
 - ▶ Analyses > Exploration > Scatterplot
- ▶ Move Gender to the x-axis and Australianness to the y-axis
 - ▶ Note, it is easier to see what direction the relationship is if you add a linear regression line (regression line > linear)

What is the relationship between Gender and Australianness?

- ▶ It is possible that males are rating themselves slightly more Australian than females, but impossible to tell if the slope is significantly different from zero until we run more analyses
- ▶ Run a point biserial correlation
 - ▶ Analyses > Regression > Correlation Matrix > Move Gender and Australianness across
 - ▶ $r = -.04$, $p = .623$

To test whether there is a relationship between gender and how Australian people perceive themselves to be, we conducted a point biserial correlation. This revealed no significant relationship ($r_{pb} = -.04$, $p = .623$). Although females were slightly more likely to perceive themselves to be less Australian than males, the result is likely to have come about by chance.

Types of correlation

► Interval/ratio x interval/ratio

	Nominal	Ordinal	Interval/Ratio
Nominal	Chi-squared (χ^2), Phi or Cramer's V, Clustered bar chart	← Treat as for	Point bi-serial correlation (r_{pb}), Scatterplot, bar chart, or error-bar chart
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Interval/ Ratio			Pearson's Product-moment correlation (r), Scatterplot

- ▶ What is the relationship between gender role and Australianness?
 - ▶ Frequencies
 - ▶ Scatterplot
 - ▶ Pearson's correlation

Univariate frequencies and bar charts

- ▶ Analyses > Exploration > Descriptives
- ▶ Move Gender role and Australianness into the variables box
- ▶ Check measurement type. Both variables should be continuous
- ▶ From the statistics menu, request mean, standard deviation, skewness and kurtosis
- ▶ From the plots menu, request Histogram

Generate scatterplots

- ▶ We can now graph the correlation in scatr
 - ▶ Analyses > Exploration > Scatterplot
- ▶ Move Gender role to the x-axis and Australianness to the y-axis
 - ▶ Remember to request a linear regression line

What is the relationship between Gender role and Australianness?

- ▶ Run a pearson's correlation
 - ▶ Analyses > Regression > Correlation Matrix > Move Gender and Australianness across
 - ▶ $r = .12$, $p = .10$

What is the relationship between Gender role and Australianness?

- ▶ There is no relationship between gender role and Australianness ($r = .12$, $p = .10$)
 - ▶ Could a relationship between these variables depend on another variable?
 - ▶ To assess this, we need to generate a scatterplot grouped by a categorical variable to see if the relationship between gender role and Australianness depends on gender

- ▶ We can now graph the correlation in scatr
 - ▶ Analyses > Exploration > Scatterplot
- ▶ Move Gender role to the x-axis, Australianness to the y-axis and Gender to the group
 - ▶ Remember to request a linear regression line
 - ▶ To follow up on the results we see, we need to run correlations between gender role and australianness separately for males and females
 - ▶ In Jamovi, we do this by filtering out rows (participants) we don't want to include in our analysis

- ▶ Variables > Filters > = Gender == 'male'
 - ▶ By filtering to select participants only when Gender == 'male', females will be deselected
- ▶ Check your data - males should have a green tick by them (selected), and females should have a red cross (deselected)
- ▶ Now the correlation between Gender role and Australianness will update - it will give you the correlation between these variables for males only
- ▶ Change the filter to select females instead of males (Variables > Filters > = Gender == 'female')
- ▶ The correlation matrix will automatically update to only give you the correlation between Gender role and Australianness for females

Does the relationship between Gender role and Australianness, depend on Gender?

- ▶ Yes. For females, the more they identify as feminine (and less masculine), the more Australian they feel
- ▶ For males, the more they identify as masculine the more Australian they feel

- ▶ What is an outlier?
- ▶ How can this affect a correlation?
- ▶ Have a play on [GeoGebra - Correlation and regression](#), to test the effect of outliers

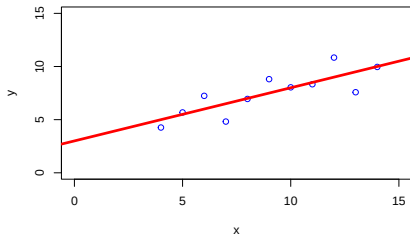
Visualising correlations is important

Anscombe dataset (statistics)

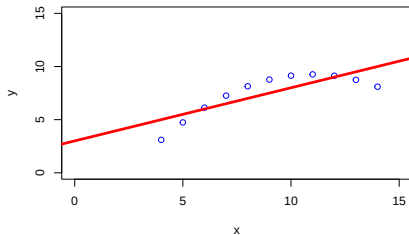
##	Mean X	Variance X	Mean Y	Variance Y	Correlation of XY
## Group 1	9	11	7.500909	4.127269	0.82
## Group 2	9	11	7.500909	4.127629	0.82
## Group 3	9	11	7.500000	4.122620	0.82
## Group 4	9	11	7.500909	4.123249	0.82

Anscombe dataset (graphs)

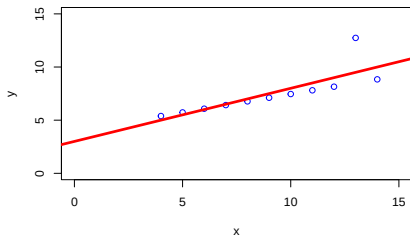
Distribution for Group 1



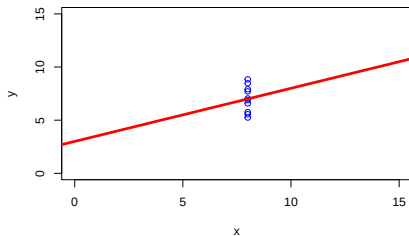
Distribution for Group 2



Distribution for Group 3



Distribution for Group 4



Questions?

- ▶ You can ask questions any time on the discussion forums.
 - ▶ Jamovi
 - ▶ Tutorial